

BUILD MANUAL · REVISION B · 30 SEPTEMBER 2026

Tomb of Cyrus, Pasargadae

A glueless, self-locking replica for 3 mm plywood, cut on the xTool M2 (10 W) from two A4 sheets. Every joint is a friction fit or a lock, and every slot is hidden in the finished model.



Height	Base	Parts	Material	Machine
122 mm	155 × 143 mm	48 + fit coupon	2 × A4, 3 mm plywood	xTool M2, 10 W blue

An interactive 3D version of this manual comes with the file set (Cyrus_Tomb_Interactive_Manual.html). It animates every step and has a part finder for both sheets.

Before you cut

1. You need two A4 sheets (210 × 297 mm) of 3 mm plywood. Basswood (linden) ply gives the cleanest stone look.
2. Also useful: a soft brush, 240-grit sandpaper, a flat board to press parts on, and calipers if you have them.
3. No glue anywhere. Every joint is a friction fit or a lock, and every slot is hidden once the model is built.
4. Finished model: 122 mm tall on a 155 × 143 mm base, 48 parts plus a fit coupon and key.
5. Sort into trays by tier: 1, 2, 3, 4, 5, 6, cross, chamber, roof. The IDs are engraved on hidden tabs and bands.
6. Rings nest inside each other on the sheets: R2, R4, R6 on sheet A and R1, R3, R5 on sheet B.
7. Rule for every part: the engraved stone face points outward or up.

CHECK

The file `Cyrus_Tomb_122mm_glueless_M2_10W_2xA4.xs` opens in xTool Studio and shows three canvases.

Machine setup

1. Select the M2, the blue 10 W laser and the flat-surface mode. The file already sets 3.0 mm thickness.
2. Leave the processing order on Auto. xTool Studio then engraves first and cuts inner shapes before the shapes around them.
3. Every part is one red cut path (outline plus its slots). Leave the green A4 outline alone: it is excluded from processing.
4. The engraving layers are starting values for untested plywood. The fit coupon checks all four of them.

TIP

Place the sheet so the green A4 outline sits on the real sheet in the camera view. Do not scale the design.

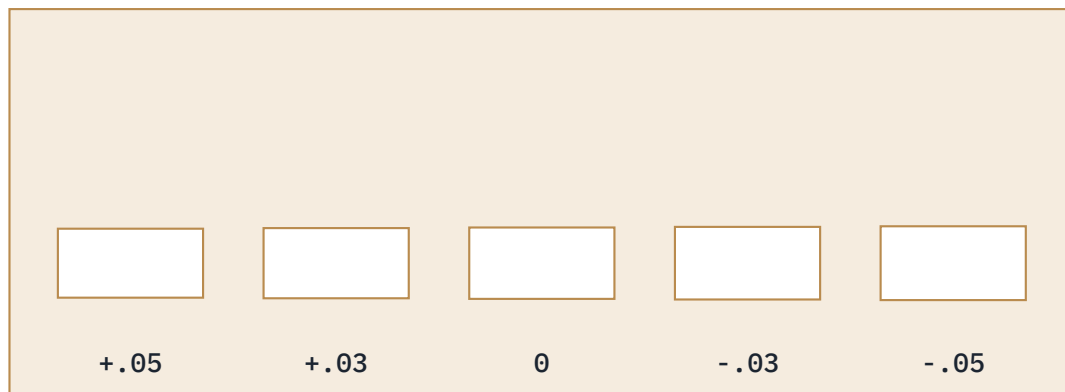
Layer	Operation	Power	Speed	Passes	Used for
1 ENGRAVE light shading	Engrave, fill 100 lines/cm	12 %	160 mm/s	1	Mouldings, rosette, door recess, chipped arrises
2 ENGRAVE dark	Engrave, fill 100 lines/cm	30 %	100 mm/s	1	Passage interior, robbed clamp sockets
3 SCORE stone joints	Score	25 %	90 mm/s	1	Block and course joints
4 SCORE detail	Score	12 %	140 mm/s	1	Cracks, moulding lines, rough foundation, part IDs
5 CUT	Cut, air assist 100 %	100 %	5 mm/s	1	Every outline with its slots, one path per part
A4 sheet edge	Not processed	–	–	–	Alignment only

Canvas 1 holds the fit coupon, canvas 2 sheet A, canvas 3 sheet B. Place each sheet so the green A4 outline matches it in the camera view, and never scale the design. Keep the processing order on Auto.

Cut and read the fit coupon

1. Put sheet A in the machine. Open canvas 1 (FIT COUPON) and process it. It sits where it belongs on sheet A.
2. Lift out the coupon and the KEY. Push the KEY into each of the five slots with your thumb.
3. Choose the narrowest slot the KEY enters fully and that still holds the KEY when you turn the coupon upside down.
4. Read the number engraved under that slot. On canvases 2 and 3, open the 5 CUT layer, switch on Kerf offset and enter that number (for example +0.03).
5. Look at the engraving swatches: light tone, dark fill, joint line, crack line. Adjust a layer power only if a swatch is burnt or invisible.

Coupon at 3 : 1. Tighter ← slots → looser



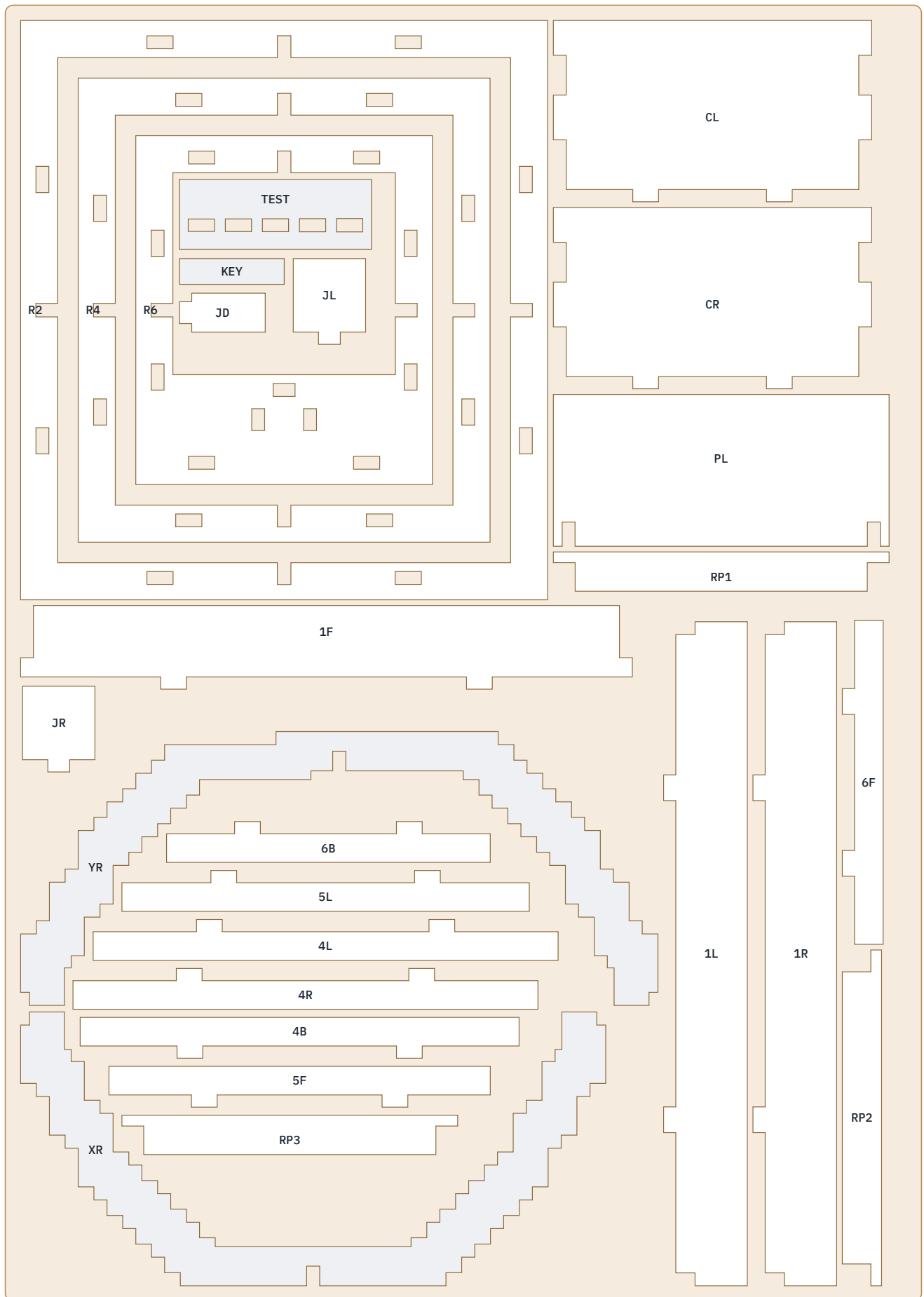
Best slot	Kerf offset on layer 5 CUT	Meaning
+.05	+0.05	Thin sheet or wide kerf: tighten all joints
+.03	+0.03	A little tighter
0	off	The design value; change nothing
-.03	-0.03	A little looser
-.05	-0.05	Thick sheet: loosen all joints

CHECK Slot marked 0 is right: leave kerf offset off. Tighter slot fits best: positive number. Looser slot: negative number.

NOTE If even the tightest slot (+.05) is loose, enter +0.07. If even the loosest (-.05) is too tight, enter -0.07.

A positive offset moves the beam outward from every shape: outlines grow and slots shrink, so every joint gets tighter. The same value applies to all 48 parts, so one number tunes the whole kit. Leave the offset off for the coupon itself.

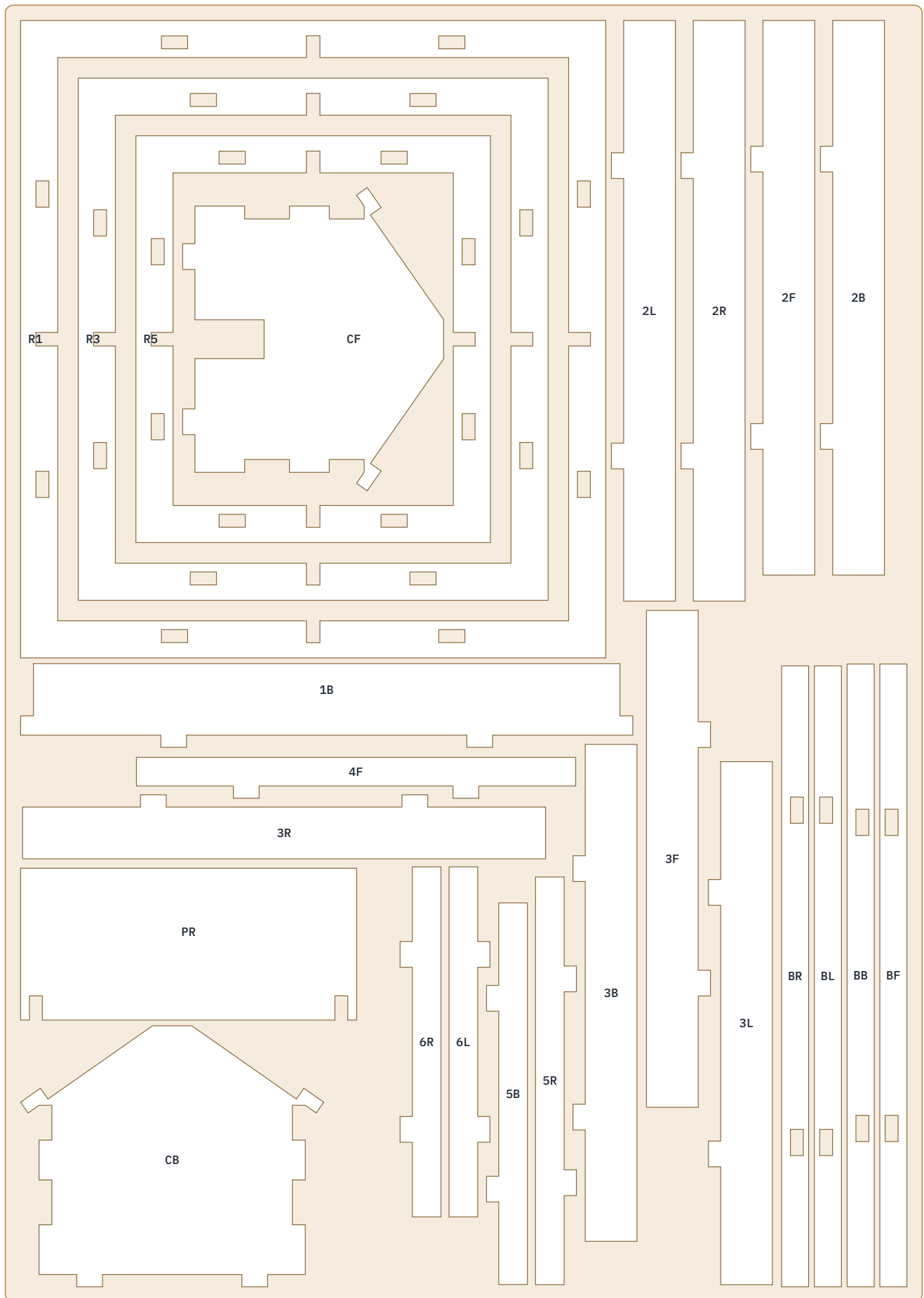
Sheet A · canvas 2



Shown at 82 % of full size. Grey parts are hidden inside the model. IDs are engraved on hidden tabs and bands.

122 mm · 2 × A4 · xTool M2 10 W · Cyrus_Tomb_122mm_glueless_M2_10W_2x4A.xs

Sheet B · canvas 3



Shown at 82 % of full size. Grey parts are hidden inside the model. IDs are engraved on hidden tabs and bands.

122 mm · 2 × A4 · xTool M2 10 W · Cyrus_Tomb_122mm_glueless_M2_10W_2x4A4.xs

Cut the sheets and sort the parts

1. Leave sheet A in place and process canvas 2 (SHEET A). Then load sheet B and process canvas 3 (SHEET B).
2. Before lifting a sheet, press gently on a corner part. If it does not drop out, run the 5 CUT layer once more on that canvas (select it, then process).
3. Keep the sheet flat until the job ends. Moving it between engraving and cutting shifts the stone joints.

TIP Start value for cutting 3 mm basswood: 100 % power, 5 mm/s, 1 pass, full air. If it does not cut through, use 4 mm/s.

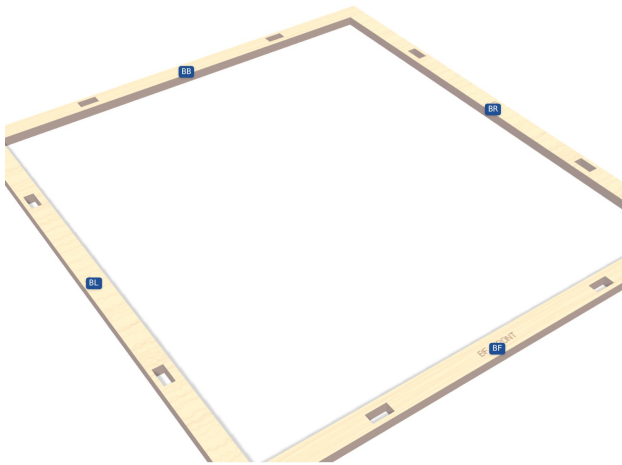
4. Pop the parts out from the back. Brush off soot; do not sand the engraved faces.
5. Sort into trays by tier: 1, 2, 3, 4, 5, 6, cross, chamber, roof. The IDs are engraved on hidden tabs and bands.
6. Rings nest inside each other on the sheets: R2, R4, R6 on sheet A and R1, R3, R5 on sheet B.
7. Rule for every part: the engraved stone face points outward or up.

ID	Part	Sheet	Step
1F	Tier 1 front riser	A	7
1B	Tier 1 back riser	B	7
1L	Tier 1 left riser	A	8
1R	Tier 1 right riser	A	8
2F	Tier 2 front riser	B	11
2B	Tier 2 back riser	B	11
2L	Tier 2 left riser	B	11
2R	Tier 2 right riser	B	11
3F	Tier 3 front riser	B	13
3B	Tier 3 back riser	B	13
3L	Tier 3 left riser	B	13
3R	Tier 3 right riser	B	13
4F	Tier 4 front riser	B	14
4B	Tier 4 back riser	A	14
4L	Tier 4 left riser	A	14
4R	Tier 4 right riser	A	14
5F	Tier 5 front riser	A	15
5B	Tier 5 back riser	B	15
5L	Tier 5 left riser	A	15
5R	Tier 5 right riser	B	15
6F	Tier 6 front riser	A	16
6B	Tier 6 back riser	A	16
6L	Tier 6 left riser	B	16
6R	Tier 6 right riser	B	16

ID	Part	Sheet	Step
BF	Foundation bar front	B	6
BB	Foundation bar back	B	6
BL	Foundation bar left	B	6
BR	Foundation bar right	B	6
R1	Tier 1 tread ring	B	11
R2	Tier 2 tread ring	A	13
R3	Tier 3 tread ring	B	14
R4	Tier 4 tread ring	A	15
R5	Tier 5 tread ring	B	16
R6	Deck (tier 6 top)	A	17
CF	Front gable with doorway	B	20
CB	Rear gable	B	20
CL	Left side wall	A	20
CR	Right side wall	A	20
JL	Door jamb left	A	18
JR	Door jamb right	A	18
JD	Dark passage back	A	19
XR	Cross rib (left-right)	A	9
YR	Cross rib (front-back)	A	9
PL	Roof slab left	A	23
PR	Roof slab right	B	24
RP1	Ridge plate (lost-capstone bed)	A	22
RP2	Ridge plate (lost-capstone bed)	A	22
RP3	Ridge plate (lost-capstone bed)	A	22

STEP 6 OF 25 · PLINTH

Lay out the foundation bars



BF BB BL BR

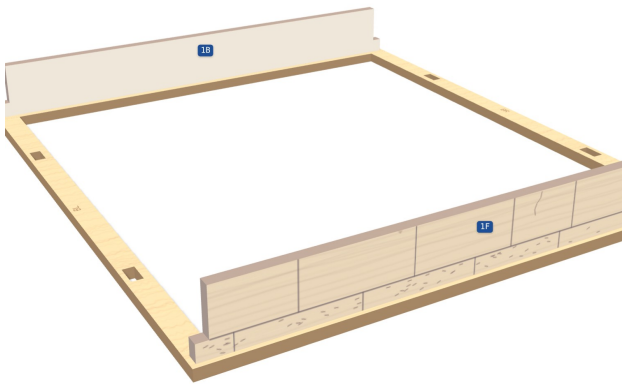
1. Place BF at the front and BB at the back, engraved side up (the text FRONT on BF faces you).
2. Put BL on the left and BR on the right, between BF and BB, so the four bars form a closed rectangle.

CHECK

Outer size about 154.6 × 142.6 mm. The eight slots line up with the wall positions.

STEP 7 OF 25 · PLINTH

Tier 1: front and back walls



1F 1B

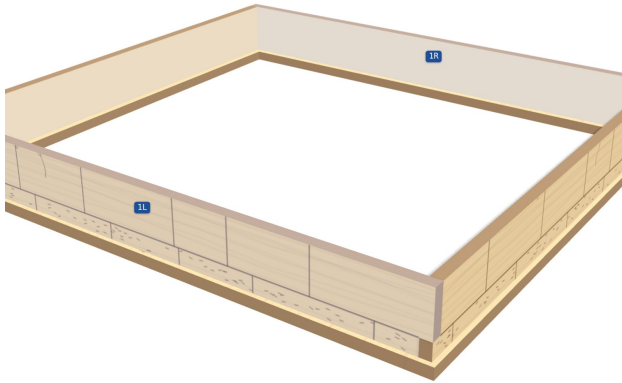
1. Hold 1F with the stone face out and the rough lower course down. Press its two tabs straight down into the slots of BF.
2. Do the same with 1B on BB.

CHECK

Both walls stand vertical and sit flat on the bars. At each end, the lower course sticks out as a short finger.

STEP 8 OF 25 · PLINTH

Tier 1: side walls

**1L 1R**

1. Lower 1L straight down on the left. Its upper corner fingers drop onto the lower fingers of 1F and 1B, and its tabs enter BL.
2. Repeat with 1R on the right.

CHECK

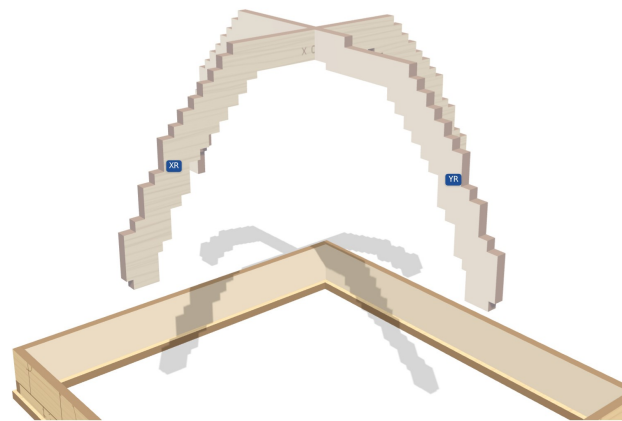
The corners interlock like real quoins: the lower course belongs to the front wall, the upper course to the side wall.

TIP

If a finger binds, check that the wall is not tilted. Never force a corner: flex the wall slightly and press again.

STEP 9 OF 25 · PLINTH

Build the hidden cross

**XR YR**

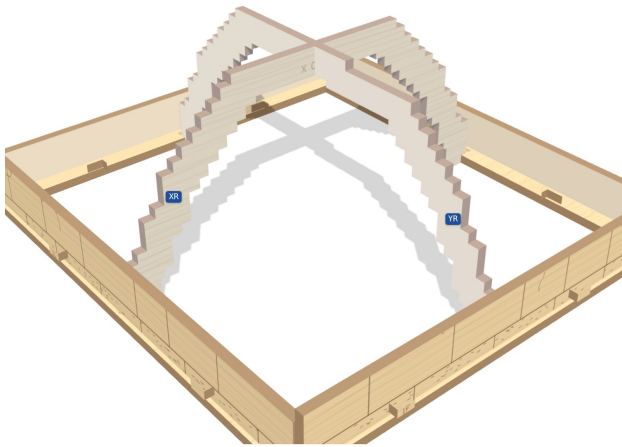
1. Hold XR (X CROSS) upright. Slide YR down onto it so the two centre slots interlock. The arrow on YR (< FRONT) points to the front.
2. The cross carries every ring and centres every tier automatically. It is invisible once built.

CHECK

Both ribs are square to each other and their top edges are flush.

STEP 10 OF 25 • PLINTH

Set the cross into tier 1



XR YR

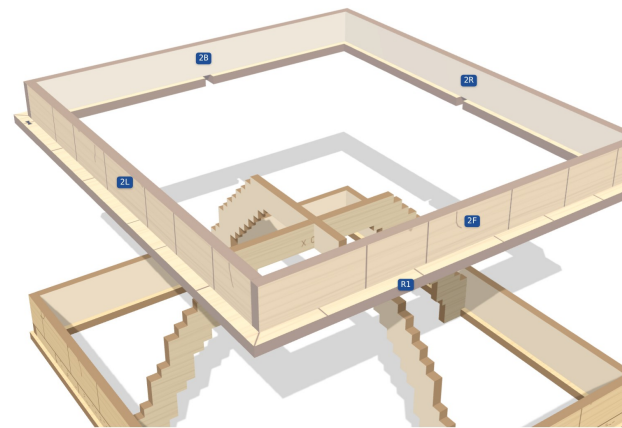
1. Lower the cross into the tier-1 box, arrow to the front.
2. The feet stand on the table just inside the foundation bars. The rib ends touch the middle of each wall.

CHECK

The cross stands upright without rocking and does not push the walls outward.

STEP 11 OF 25 • PLINTH

Module 2 on the bench



R1 2F 2B 2L 2R

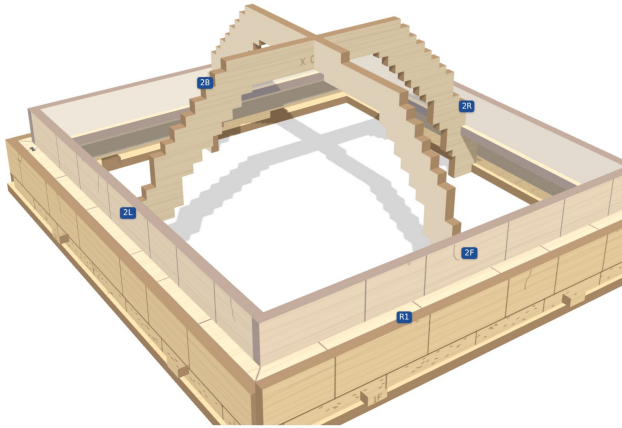
1. Lay ring R1 flat, engraved side up, with R1 FRONT toward you.
2. Press 2F and 2B into the slots on the front and back. Then press 2L and 2R in between them.
3. Work on a flat board and push each wall with both thumbs.

CHECK

All four walls stand on the ring, the corners close, and the stone joints on the ring line up with the joints on the walls.

STEP 12 OF 25 · PLINTH

Lower module 2 onto tier 1



R1 2F 2B 2L 2R

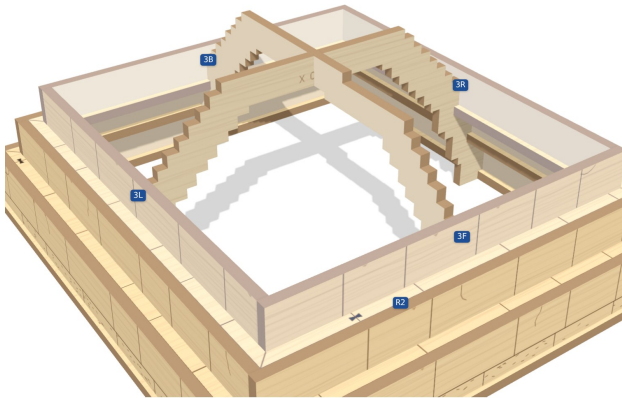
1. Hold the module by its walls and lower it over the cross, R1 FRONT to the front.
2. The four notches in R1 slide down the ribs. The ring drops inside the tier-1 walls and rests on the cross.

CHECK

R1 is flush with the top edges of the tier-1 walls on all four sides. The step width is even all round.

STEP 13 OF 25 · PLINTH

Module 3: ring R2 + tier 3 walls



R2 3F 3B 3L 3R

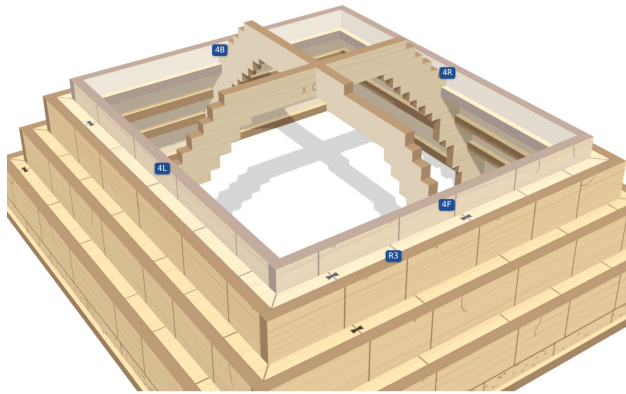
1. On the bench: ring R2 engraved side up, R2 FRONT toward you. Press in 3F and 3B, then 3L and 3R.
2. Lower the module over the cross until R2 drops inside the tier-2 walls.

CHECK

R2 sits flush with the tier-2 walls.

STEP 14 OF 25 · PLINTH

Module 4: ring R3 + tier 4 walls



R3 4F 4B 4L 4R

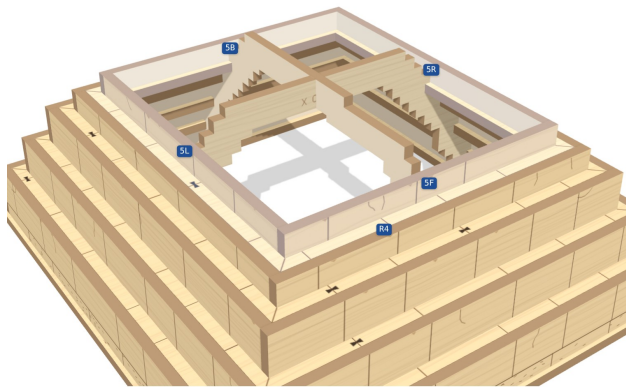
1. On the bench: ring R3 engraved side up, R3 FRONT toward you. Press in 4F and 4B, then 4L and 4R.
2. Lower the module over the cross until R3 drops inside the tier-3 walls.

CHECK

R3 sits flush with the tier-3 walls. Tiers 4 to 6 are low: hold them by the ring, not the thin walls.

STEP 15 OF 25 · PLINTH

Module 5: ring R4 + tier 5 walls



R4 5F 5B 5L 5R

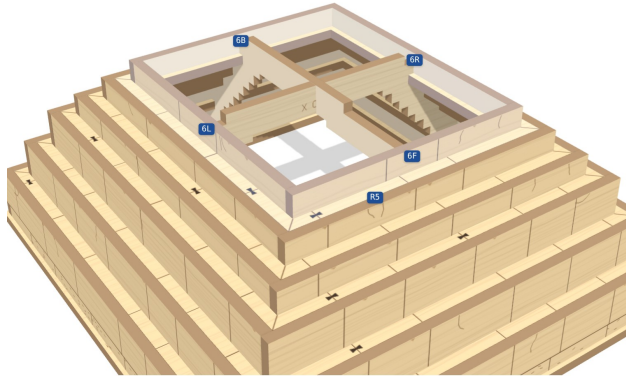
1. On the bench: ring R4 engraved side up, R4 FRONT toward you. Press in 5F and 5B, then 5L and 5R.
2. Lower the module over the cross until R4 drops inside the tier-4 walls.

CHECK

R4 sits flush with the tier-4 walls. Tiers 4 to 6 are low: hold them by the ring, not the thin walls.

STEP 16 OF 25 • PLINTH

Module 6: ring R5 + tier 6 walls



R5 6F 6B 6L 6R

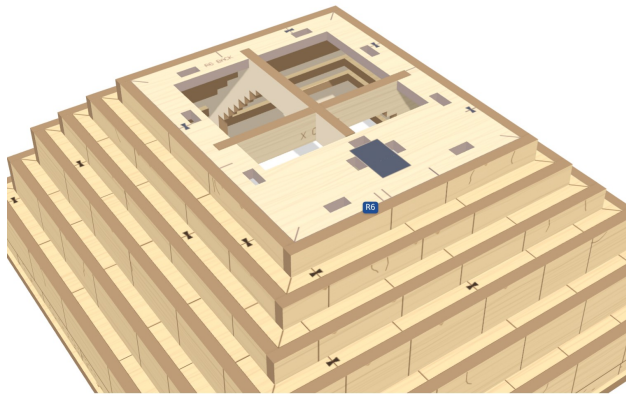
1. On the bench: ring R5 engraved side up, R5 FRONT toward you. Press in 6F and 6B, then 6L and 6R.
2. Lower the module over the cross until R5 drops inside the tier-5 walls.

CHECK

R5 sits flush with the tier-5 walls. Tiers 4 to 6 are low: hold them by the ring, not the thin walls.

STEP 17 OF 25 • PLINTH

Drop in the deck



R6

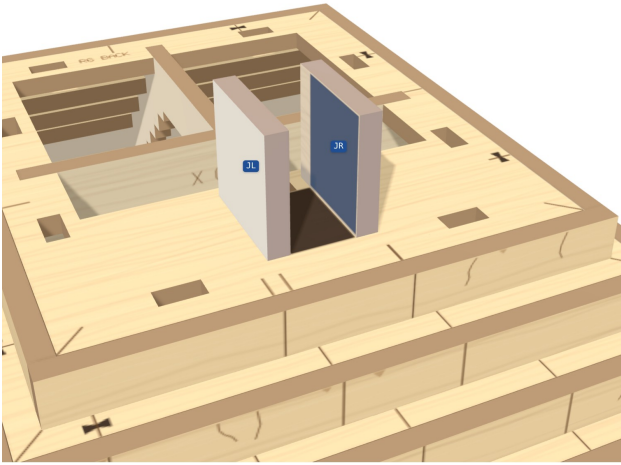
1. Place R6 engraved side up. The wide band with the dark patch goes to the front: that is the floor of the doorway.
2. It drops into the tier-6 box and rests on the cross.

CHECK

Plinth height about 62.8 mm. The top is level; press each corner and nothing rocks.

STEP 18 OF 25 • CHAMBER

Door jambs



JL JR

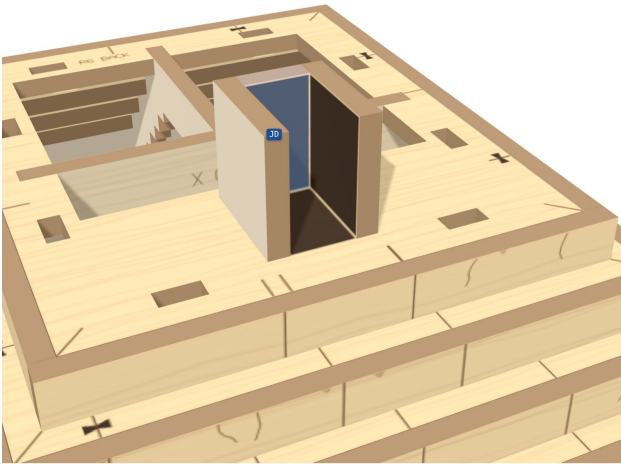
1. Press JL and JR into the two small slots near the front of the deck.
2. Their dark faces look at each other: they form the walls of the passage behind the doorway.

CHECK

The jambs are parallel and about 9 mm apart, the width of the doorway.

STEP 19 OF 25 • CHAMBER

Dark passage back



JD

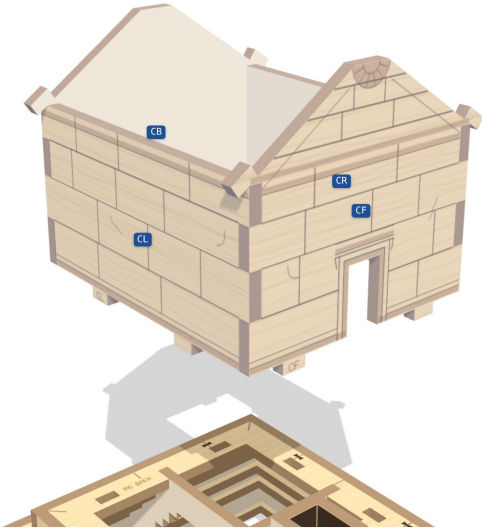
1. Slide JD down between the back ends of the jambs, dark face toward the door. Its tab enters the deck.

CHECK

Looking through the future doorway you see only dark faces.

STEP 20 OF 25 • CHAMBER

Chamber walls on the bench



CF CL CR CB

1. Stand CF (the gable with the doorway) on the bench, stone face out.
2. Slide CL and CR sideways into its corners so the corner fingers interleave course by course.
3. Slide CB onto the back ends of CL and CR the same way.

CHECK

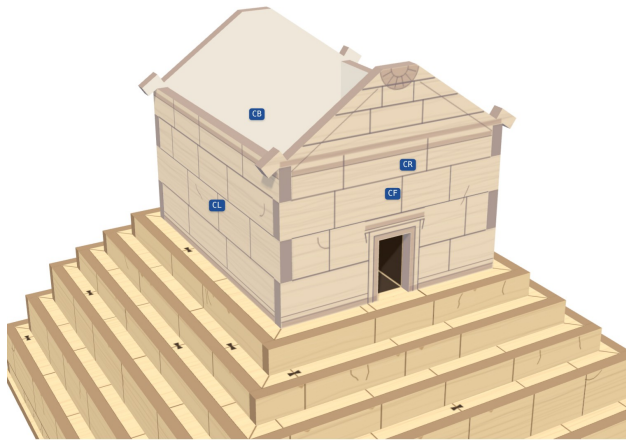
Four closed corners with alternating quoins.
The box stands square.

TIP

Slide, do not lift: these corners have four fingers each, so they only go together horizontally.

STEP 21 OF 25 • CHAMBER

Set the chamber on the deck



CF CL CR CB

1. Lower the box straight down around the door jambs, doorway to the front.
2. Press evenly until all eight tabs sit in the deck.

CHECK

The doorway lines up with the passage and the chamber is centred on the deck with an even ledge all round.

STEP 22 OF 25 · ROOF

Ridge plates



RP1 RP2 RP3

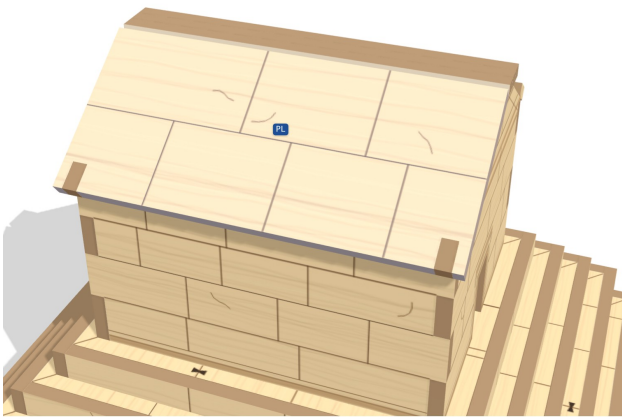
1. Set the three ridge plates side by side on the flat tops of the two gables.
2. The slots in their lower edge straddle the gables. Together they form the flat top where the capstone is missing.

CHECK

The three top edges are flush and the block overhangs both gables by the same amount.

STEP 23 OF 25 · ROOF

Left roof slab



PL

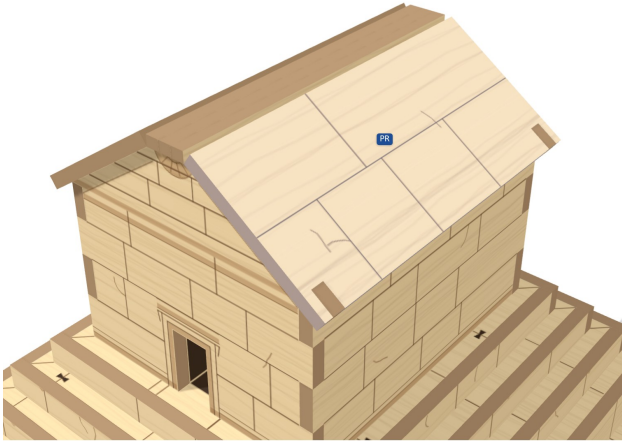
1. Hold a slab with the engraved roof stones up. Place its lower edge first: the two notches drop over the small teeth at the gable corners.
2. Lay the slab down onto the gables. Its upper edge rests against the ridge plates.

CHECK

The slab lies flat on both gables and overhangs them equally.

STEP 24 OF 25 · ROOF

Right roof slab



PR

1. Fit the second slab the same way.
2. The two slabs now hold the ridge plates between them.

CHECK

The roof is symmetrical and the half rosette sits right under the flat top on the front gable.

STEP 25 OF 25 · ROOF

Final check



1. Height 122.2 mm to the flat top. Six receding tiers, a gabled chamber and a dark doorway.
2. Press each base corner lightly: the model must not rock. No slot or tab is visible from the outside.
3. To move the model, lift it by the foundation, never by the roof.

CHECK

Optional: a very thin wash of warm grey water colour on scrap first, then on the model, gives a limestone tone and keeps the joints readable.

Dimensions, checks and sources

Feature	Monument	Model	Model at full size
Total height (flat top, capstone missing)	≈ 11.0 m (UNESCO)	122.2 mm	10.72 m
Plinth base (L × W)	13.35 × 12.30 m	152.2 × 140.2 mm	13.35 × 12.30 m
Tier heights	1.70 / 1.04 / 1.04 / 0.575 × 3 m	19.4 / 11.9 / 11.9 / 6.6 / 6.6 / 6.6 mm	same
Plinth height	5.50 m	62.8 mm	5.50 m
Cella base (L × W)	6.40 × 5.35 m	73.0 × 61.0 mm	6.40 × 5.35 m
Doorway (w × h)	0.78 × 1.39 m	8.9 × 15.8 mm	0.78 × 1.39 m
Roof pitch	steep (not published)	35.0°	reconstruction

Simulated before release

- 3D collision test of all 48 parts in their built positions: 0 overlaps at nominal fit.
- Every one of the 59 slots is filled by a tab of the right part, and every tab has a slot.
- Ray test from 126 viewing directions around and above the model: no slot or notch can be seen. The only openings face the table.
- Every assembly move simulated in 0.5 mm increments over 40 mm of travel: no part hits another on its way in.
- Every part touches its neighbours; nothing floats.
- Both sheets: at least 3.4 mm from the sheet edge and 1.3 mm between parts, after kerf compensation.
- The .xs file checked against real xTool Studio M2 exports: file layout, display keys, profiles, bindings, closed cut paths. 0 errors.
- 45 press-fit contacts, each with 0.05 mm designed interference for a glueless hold.

Only a physical test can confirm

- Your plywood thickness and the real kerf of your M2: the coupon covers both.
- Cut-through at 5 mm/s and the darkness of each engraving layer on your batch of wood.
- The .xs file follows the structure of real xTool Studio M2 exports and passed every structural check, but it has not been opened in xTool Studio or cut on a machine.
- Set-backs, course heights and the 35° roof pitch are reconstructions where no survey value is published.

Sources

- UNESCO World Heritage nomination 1106, Pasargadae: whc.unesco.org/uploads/nominations/1106.pdf
- Encyclopaedia Iranica, “Cyrus v. The Tomb of Cyrus”: iranicaonline.org/articles/cyrus-v-tomb
- Encyclopaedia Iranica, “Pasargadae”: iranicaonline.org/articles/pasargadae
- xTool M2 FAQ (support.xtool.com/article/3324) and kerf offset (support.xtool.com/article/2426)